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An Improved Physics-guided Interpretable Inception Network and Its Application in Bearing Fault Diagnosis

Jingyuan Wu, Diwang Ruan, and Yiliang Qian

***Abstract*—In industrial applications, obtaining sufficient fault samples is challenging due to factors like sensor installation difficulties and limited fault occurrence. Consequently, small sample size becomes an inevitable obstacle for data-driven fault diagnosis. The Inception network demonstrates strong performance in bearing fault diagnosis, particularly under small sample conditions owing to its structural advantages. However, the commonly used Inception network suffers from poor interpretability and is prone to overfitting. To address these issues, this paper proposes a physics-guided Inception network (PG-Inception), leveraging the physical characteristics of bearing vibration signals to guide the design of its parameters and structure. First, the input length and size are designed based on rules derived from the fault characteristic frequency and shaft rotation frequency. Next, by analyzing the acceleration response within a single fault period and considering possible fault types, the convolution kernel size and the number of parallel branches in PG-Inception are determined. Finally, the effectiveness of PG-Inception is validated using the Case Western Reserve University (CWRU) and the Paderborn University (PU) bearing dataset. The experimental results indicate that under small sample conditions, PG-Inception achieves significantly higher accuracy, faster convergence, and stronger stability compared to the baseline Inception, validating the effectiveness and superiority of the proposed method.**

***Index Terms*—Bearing, Parameter design, Physics-guided Inception (PG-Inception), Interpretability, Fault diagnosis, Small samples.**

I. Introduction

BEARINGS are widely used in various industrial equipment, and their operational status directly influences the performance and reliability of the entire system. Bearing failures not only degrade equipment performance but can also lead to severe damage and safety accidents [1]. Therefore, bearing fault diagnosis has become a crucial aspect of mechanical equipment maintenance, providing significant research and application value. Accurate and timely detection of bearing faults can effectively reduce equipment downtime and maintenance costs while enhancing production efficiency and safety [2], [3].

With the rapid advancement of technologies like machine learning, data-driven intelligent fault diagnosis methods have been increasingly adopted. However, these methods are highly data-dependent, requiring substantial amounts of effective data from various operating states to train accurate fault diagnosis models. In industrial production environments, challenges such as sensor installation difficulties and limited fault state duration hinder the collection of sufficient fault data, complicating the training of high-accuracy fault diagnosis models. Zhang et al. demonstrated that when using a fault diagnosis model based on the wide first-layer kernel deep convolutional neural network (WDCNN) algorithm, the accuracy declined from 94.32% to 82.80% as the sample size decreased from 200 to 60. Under similar conditions, the accuracy of a fault diagnosis model based on the support vector machine (SVM) algorithm dropped from 38.67% to 18.93% [4]. Therefore, developing methods to effectively train fault diagnosis models under small sample conditions is crucial for advancing the application of intelligent fault diagnosis in real-world production.

There are two approaches to addressing the small sample problem. One approach relies on algorithmic models, utilizing deep learning techniques to extract as much information as possible from the available samples. Liu et al. proposed a model based on the CNN-BiLSTM algorithm, which combines the strengths of convolutional neural networks (CNN) and bidirectional long short-term memory (BiLSTM) algorithms. This model simultaneously extracts both spatial and temporal features, improving the accuracy for 100 and 500 samples from 61% and 76% to 96% and 98%, respectively [5]. Xie et al. employed a feature extraction module based on multi-scale feature fusion and a metric learning module to input vibration signals into sample pairs. This method can increase the number of learning tasks and improve the fault diagnosis capability of the network under small sample conditions [6]. To enhance practical application capabilities and improve model operation speed in real-world scenarios, Lee et al. proposed a lightweight WDCNN model. While improving the accuracy of small sample fault diagnosis, it reduced the number of model parameters and enhanced the operation speed [7]. The other approach is data-based, which primarily expands the number of samples through data enhancement, pseudo-sample generation, and other techniques to provide more samples for training fault diagnosis models. Ruan et al. used an exponential function to fit the health index curve and modeled the distribution of the coefficients. Finally, they employed the Monte Carlo method to resample the parameters and expand the health index dataset, improving the accuracy and stability of remaining useful life (RUL) predictions [8]. Ruan et al. applied the collaborative optimization of CNN and generative adversarial network (GAN) to use the fault diagnosis results from CNN and the envelope spectrum error of fault characteristic frequency (FCF) as two correction terms in the GAN generator loss function. The results showed that

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this method contained more fault information than an ordinary GAN, increasing fault diagnosis accuracy from 88% and 68% to 90% and 88% [9]. Cheng et al. proposed a multimodal few-shot learning (MMFSL) method that used two channels to expand time-series and image data separately, increasing the fault detection accuracy from 80.9% to 97.5% [10].

In 2015, Szegedy et al. proposed the Inception network, which achieves multi-scale feature extraction by connecting multiple branches in parallel. Its main feature is the improved utilization of computing resources within the network [11], showing strong performance under small sample conditions [12]. Following its introduction, Szegedy updated the structure three times in three years. Inception-v2 was proposed in 2015, incorporating Batch Normalization (BN) technology. A BN layer was added after the convolution of each branch to accelerate training and improve model convergence [13]. Inception-v3 was proposed in 2016, introducing an Auxiliary Classifier (AC) that provided additional supervision signals during training to mitigate the gradient vanishing problem, accelerate convergence, and improve the accuracy and generalization ability of the model [14]. Inception-v4 was proposed in 2017, incorporating additional optimization techniques, such as residual connection and layer-by-layer normalization, to further improve the performance and training speed of the model. Inception-v4 also uses data enhancement strategies and regularization techniques to reduce the risk of overfitting and improve the generalization ability of the model [15]. At the same time, Szegedy also proposed the Inception-ResNet structure, which combines the multi-scale feature extraction capability of the Inception module with the residual connection of ResNet to solve the gradient vanishing problem in deep networks and improve training speed. The Inception-ResNet structure has been widely applied to the field of bearing fault diagnosis, yielding excellent results [15]. Liu et al. and Tian et al. enhanced the model's feature expression ability by integrating transfer learning with the Inception-ResNet network, improving the accuracy of fault diagnosis under small sample conditions [16], [17]. Wei et al. combined the Inception network with BiLSTM and applied it to tool fault state recognition under small sample conditions, increasing accuracy from 78.5% to 87.5% [18]. Zhang et al. combined model-agnostic meta-learning (MAML) with Inception, improving fault diagnosis accuracy under small sample conditions by 25% [19].

The interpretability of deep learning algorithms is closely related to the credibility of the results, and its difficulty stems from the highly nonlinear end-to-end mapping of the algorithm. Therefore, studying the interpretability of deep learning algorithms can improve the understanding of the algorithm and the trust in the results [20]. At present, most studies focus on improving interpretability by introducing constraints into the loss function. Zhang et al. designed a loss function based on physical information to evaluate the gap between the output of the deep learning model and the update of the finite element model, thereby enhancing the interpretability of the results [21]. To improve the accuracy of RUL prediction for bearings, Chen et al. incorporated loss function constraints into a recurrent neural network (RNN), which resulted in improvements in both prediction accuracy and interpretability [22]. Different from the above methods, Ruan et al. used physical information to guide the design of CNN model parameters. Without integrating other algorithms, they forward-designed the deep learning model according to the characteristics of the input information, thereby improving the efficiency and interpretability of the model [23].

Based on the above analysis, this paper proposes an approach grounded in the fault mechanism, using an analysis of the bearing vibration signal to guide the design of the input size, the number and size of parallel branches in the Inception network. This approach enhances the interpretability of the network compared to the baseline Inception. The method proposed in this paper directly determines model parameters such as input length and convolution kernel width based on the physical characteristics of bearing vibration signals, including FCFs and signal attenuation characteristics. By designing the model based on physical principles, our approach reduces the reliance on extensive hyperparameter tuning and enhances the model interpretability through a forward design mechanism. This differs from the methods proposed by Shen et al. and Jiang et al. The former optimizes the model by introducing physical knowledge into the loss function of model training, while the latter designs a feature extractor combined with a GAN [24], [25].

From the perspective of physics-informed machine learning (PIML), the proposed PG-Inception model can be viewed as a structure-level physics-guided approach. Recent PIML studies in Prognostics and Health Management (PHM), such as those by Wen et al., who fused empirical and physical degradation models for lithium-ion batteries [26], and by Barimah et al., who developed a scalable PINN-based digital twin for multi-component hydraulic systems [27], typically place the physical model in the loss layer during training and achieve backward optimization. In contrast, PG-Inception uses the physical model to guide the design of the data-driven network before training, achieving alignment between the model architecture and the data at the structural and parameter levels. This represents a forward design strategy. Therefore, the proposed method can be viewed as a new implementation framework of PINN, integrating the strengths of both model-driven and data-driven approaches and thereby enabling physical interpretability together with the adaptability of deep learning.

This paper first analyzes the physical information in the vibration signals of bearings, designs the parameter settings for the baseline Inception network, and constructs the PG-Inception network. Verification experiments are then conducted on the Case Western Reserve University (CWRU) and the Paderborn University (PU) bearing datasets, comparing the efficiency, accuracy, and performance of PG-Inception with baseline Inception under different small sample conditions. Through the analysis of the experimental process and results, the correctness of the proposed method is confirmed, enhancing the interpretability of the network. The main innovations of this paper are as follows.

1. Parameters and structural dimensions of the Inception network are designed based on information derived from

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acceleration signals, resulting in the development of the PG-Inception.

2. The efficiency and accuracy of the Inception network in tackling small sample problems have been enhanced. As the sample size decreases, the performance degradation rate of the PG-Inception network is reduced.

The rest of this paper is organized as follows. Section II introduces the physics-guided parameter design theory and discusses the design of parameters for the input sample size and parallel fault branches of the Inception network. Section III presents the experimental results, proving the effectiveness and superiority of the proposed method. Section IV concludes the paper.

II. Methodology and Implementation of Physics-Guided Inception Network

A. Introduction to Inception network

The main idea of the Inception architecture is to approximate and overlay the optimal local sparse structure of convolutional vision networks with readily available dense components [11]. As shown in Fig. 1, the structure of Inception-v1 primarily consists of parallel convolutional layers, including four branches, each connected in parallel with convolutions of different sizes. The top branch uses a filter of size (1,1), which retains the original features and reduces the number of channels, thereby decreasing computational complexity. The two middle branches are connected in parallel with filters of sizes (3,3) and (5,5), mainly for feature extraction at different receptive field sizes, enhancing the network's ability to perceive regions of varying sizes. The bottom branch is a (3,3) pooling layer, which applies max pooling to preserve key features and improve the stability of the structure. In addition, a filter of size (1,1) is connected in series in both fault branches to reduce the dimension and increase the calculation speed of the model [14].

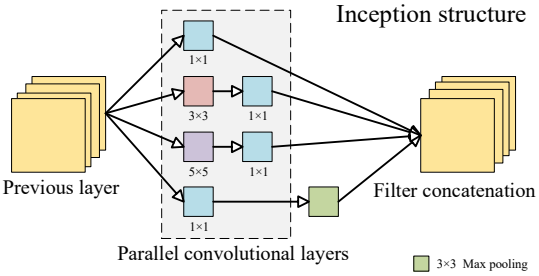


Fig. 1: Structure overview of Inception.

In general, the Inception network represents a significant advancement in deep learning due to its efficient multi-scale feature extraction capabilities and relatively low number of parameters. Although multiple versions of the original Inception network exist, its core concept remains the simultaneous learning of features at multiple scales and levels, thereby enhancing network performance. In this paper, the Inception network described above is used as the baseline Inception.

B. Physics-guided parameters design for improved Inception network

After introducing the baseline Inception, this section will present how to guide the design of Inception parameters based on physical information, including input size, parallel branch size, and structural parameters.

1) Design of input length and size: After Szegedy et al. proposed the Inception network in 2015, it has been primarily applied in computer vision, image processing, and other fields [28], [29]. For example, Xia et al. utilized the Inception network to address the significant challenge flower of classification in botany, greatly improving the classification accuracy [30]. Similarly, Wang et al. employed the Inception network for automatic feature extraction to solve lung image classification issues, achieving approximately 2% higher accuracy compared to DCNN and other methods [31]. In these cases, images are used as direct inputs to the Inception network, with the input size determined solely by the size of the image, and no specific adjustments are made to the input size. However, according to Richter et al.'s research, adjusting the input size could increase the accuracy from 84.64% to 92.79% [32]. Therefore, designing input size to better suit sample characteristics can improve network performance. This paper analyzes the physical characteristics of the bearing vibration signal, as shown in Fig. 2(a), and subsequently designs the input size. Fig. 2(b) illustrates the process of designing the input size.

Vibration signals in typical rotating machinery, such as bearings, exhibit periodic characteristics. Therefore, selecting an appropriate input size can improve the operation speed of the network and prevent the duplication of small features. Common rolling bearing faults include inner ring surface defects, outer ring surface defects, rolling element surface defects, and cage failures. To ensure that input samples contain sufficient fault information and can distinguish between different fault types, two input length restrictions are proposed based on both the time domain and frequency domain perspectives. From the perspective of the time domain, the sample length in each fault response cycle can be obtained by dividing the sampling frequency by the FCF. To ensure that the input sample length provides sufficient temporal resolution, the first criterion is defined as L_s^1. Here, sufficient resolution means that the sample contains enough sampling points to fully cover at least one complete fault cycle, thereby retaining all relevant information within the cycle. Besides separately identifying the possible FCFs, a more critical aspect of bearing fault diagnosis is distinguishing the FCF peaks from different fault positions and orders. To meet this requirement, the second condition for the sample length, denoted as L_s^2, is introduced. Its core idea is to ensure adequate frequency resolution in the spectral domain to distinguish between any two adjacent FCF peaks of different types and orders. As shown in Equations (1) and (2) [23],

$$L_s^1 = \frac{\alpha \cdot f_s}{\min(BPFI, BPFO, FTF, BSF)}, \tag{1}$$

$$L_s^2 = \frac{f_s}{\min(f_{FCF} \cdot i \neq f_{FCF} \cdot j)}, \tag{2}$$

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where f_s denotes the sampling frequency, and α ($\alpha \geq 2$) is a sampling factor defined according to Shannon sampling. BPFI (ball pass frequency inner race), BPFO (ball pass frequency outer race), FTF (fundamental train frequency), and BSF (ball spin frequency) represent the FCFs of the four fault types. These two values represent the minimum required sample lengths derived from the time and frequency domains, respectively. They ensure that each sample contains sufficient fault information to support accurate classification of fault types. Therefore, the initial sample length is determined as the maximum of L_s^1 and L_s^2. However, to facilitate comparison in later Inception classification tasks, which typically use square-shaped inputs, the final sample length L_s is set to the nearest perfect square greater than or equal to this maximum. The calculation formula is provided in Equation (3) [23]:

$$L_s = \left\lceil \sqrt{\max(L_s^1, L_s^2)} \right\rceil^2. \tag{3}$$

When reshaping the samples from one dimension to matrix input, simply cutting the samples according to mathematical requirements and stacking them row by row can result in feature loss. Since each FCF includes a component of the shaft rotation frequency f_r, the length of a single failure cycle is inversely related to f_r. The final input size is derived from Equations (4) and (5) [23],

$$W = \left\lceil \frac{f_s}{f_r} \right\rceil, \tag{4}$$

$$H = \left\lceil \frac{L_s}{W} \right\rceil, \tag{5}$$

where W and H represent the width and height of the input size, respectively.

2) Design of parallel branches: To achieve efficient feature extraction and obtain as much useful information as possible from limited samples, the structure of Inception model is designed with the following key considerations. First, it performs multiple convolution and pooling operations of varying sizes simultaneously within a single module and then connects the results. This enables the Inception model to fuse feature information from different convolution kernel sizes, enhancing the diversity and completeness of feature representation. Second, to reduce computational complexity and increase processing speed, the Inception module uses a (1,1) filter to reduce the number of channels, thereby reducing the amount of computation without losing too much information. This dimension reduction operation not only alleviates computational demands but also facilitates the introduction of nonlinear activation functions, improving the expressiveness of the model. In addition, one of the design inspirations of the Inception model is sparsity. By introducing multiple parallel branches, the network can simulate a sparse connection pattern while approximating it through dense computations. This approach retains the advantages of sparse networks while leveraging the efficiency of dense computations. Therefore, the multi-branch parallel structure and multi-scale convolution kernels are defining features that distinguish the Inception model from conventional CNN.

The method proposed in this paper designs parallel branches, which consist of two parts, as displayed in Figs. 2(c) and 2(d). First, the number of parallel branches is designed. As shown in Fig. 1, the topmost branch in an Inception module uses a (1,1) filter, while the lowest branch combines a (1,1) filter with a (3,3) pooling layer. These branches serve to preserve important feature information, enhance the model's robustness against deformations and noise, reduce dimensions, and increase operational speed, with their sizes not requiring guidance based on physical information. For convenience, the number of branches excluding these two is referred to as the fault branch number, denoted as N_b.

Essentially, the parallel structure allows convolution kernels to perceive feature maps over a broader field of view. The convolution kernels are optimized for the best perceptual field for specific faults. However, this kernel size design can accommodate only specific fault types. To achieve optimal perception for all fault types within an Inception module, the number of fault branches should equal the number of fault types, as formulated in Equation (6):

$$N_b = N_f, \tag{6}$$

where N_f represents the number of fault types. Using this method, the number of fault branches is determined. As previously analyzed, the convolution kernel size of each branch is designed to correspond to a specific fault type.

Next, the size of the convolution kernel is designed, as it determines the field of view for the extracted features. If the field of view is too small, it becomes difficult to capture large-scale features and global information, causing the model to overemphasize local features. Conversely, if the field of view is too large, the complexity and computational cost of the model increase significantly, with the risk of losing detailed information. Therefore, designing an appropriately sized convolution kernel is essential.

As presented in Fig. 2(d), the entire process is divided into four steps. Considering that the fault cycle represents a complete dynamic process triggered by a single fault event, the first step is to extract the acceleration response corresponding to one full fault cycle from the raw vibration signal, so as to capture the complete dynamic characteristics of the bearing under a single impact. To mitigate the effects of measurement noise and structural fluctuations, the second step involves applying envelope processing to the single-cycle signal, thereby enabling a more accurate characterization of the signal's attenuation trend [23]. These two steps are illustrated in Fig. 2d(1). To ensure generalizability, the design rules should be derived from analytical formulations rather than discrete measurement data. Accordingly, the third step is to fit the signal envelope using an exponential decay model. This process is formulated in Equation (7) [23]:

$$A(t) = \zeta A_0 e^{-B_\zeta t}, \tag{7}$$

where A_0 is the maximum signal amplitude within the selected fault period, B_ζ is the decaying parameter, t is time, and ζ is the amplitude decaying ratio ($0 < \zeta < 1$). The final step is to determine the convolution kernel width. Based on the exponential decay model, the time required for the signal to

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decay from A_0 to ζA_0 is calculated, and the corresponding kernel width is then obtained by incorporating the sampling frequency [23]. This process is depicted in Fig. 2d(3). The resulting width ensures that the convolution kernel adequately captures the primary dynamic content within the signal decay process, thereby enabling physics-guided feature extraction, which are calculated using Equations (8) and (9) [23],

$$t = \frac{-\ln\left(\frac{A(t)}{A_0}\right)}{B_\zeta}, \tag{8}$$

$$W_k(\zeta) = \lceil t_\zeta \cdot f_\zeta \rceil = \left\lceil \frac{-\ln \zeta}{B\zeta} \cdot f_s \right\rceil. \tag{9}$$

Then, the above three steps are repeated for the remaining two types of fault signals. In this way, three convolution kernel widths corresponding to the three fault types are obtained, as illustrated in Fig. 2d(4). This paper does not identify physical information to guide the design of the convolution kernel height. Therefore, the convolution kernel height is determined based on experimental data. As illustrated in Fig. 2c(2), the designed convolution kernel replaces the original square convolution kernel to construct the parallel branches.

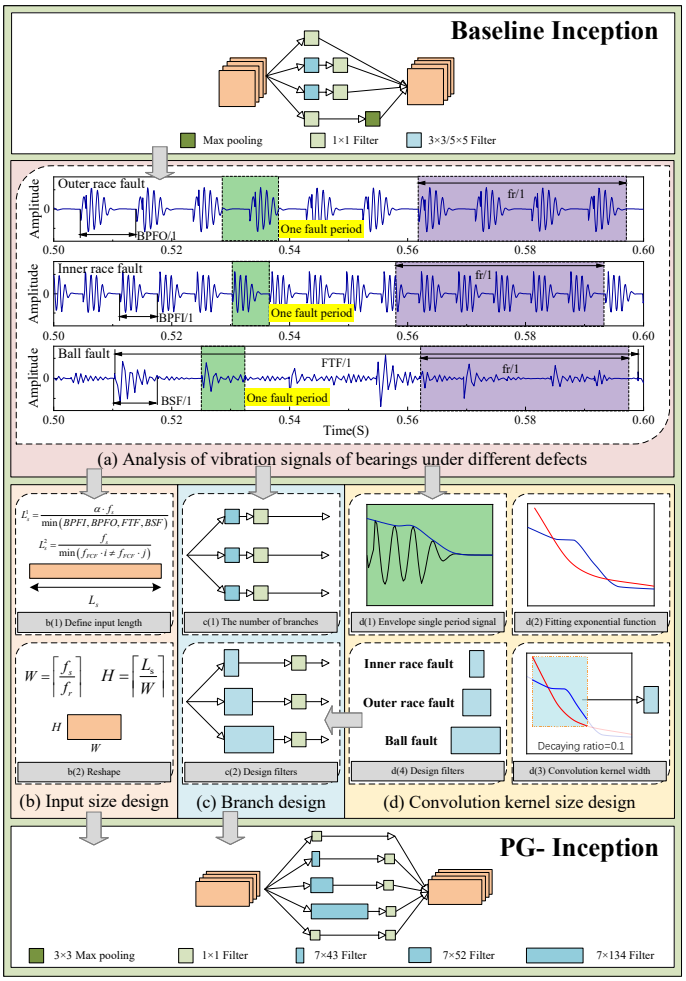


Fig. 2: Flowchart of physics-guided interpretable Inception network.

III. Experimental Verification

A. Introduction to experimental equipment and bearing datasets

The experiments utilize the CWRU and PU bearing datasets to validate the proposed method. In this section, we introduce the two experimental platforms and provide detailed information about the datasets.

The CWRU bearing experimental platform is shown in Fig. 3. It consists of a three-phase induction motor, a torque sensor, a power test sensor, and a controller. The SKF6205-2RS bearing at the drive-end is used in this study, with a sampling frequency of 12 kHz and a rotational speed of 1750 rpm. The dataset includes four states: rolling element fault, inner race fault, outer race fault, and normal condition, with corresponding labels 0 to 3. Three different fault sizes are used, including 0.007 inches, 0.014 inches, and 0.021 inches [33]. The original vibration signal is segmented into samples of 5329 data points each, resulting in a total of 11778 samples for the dataset used in this experiment. Detailed information about the original dataset is provided in Table I.

The Inception structure exhibits improved performance under certain small sample conditions due to its advantages [34], [35]. To evaluate its effectiveness, six datasets are generated by randomly sampling from the original dataset at different ratios. These experimental datasets are divided into training, test, and validation sets in a 7:2:1 ratio. It is important to note that, in this process, there is no strict requirement for maintaining the proportion of each fault type, only the total number of samples in each dataset. Detailed dataset information is presented in Table II.

The PU bearing experimental platform is shown in Fig. 4. From left to right, the platform includes a motor, torque measurement shaft, rolling bearing test module, flywheel, and load motor. The phase current signal of the drive motor and the vibration signal of the bearing box are collected. The experimental test bearing models are all 6203 rolling bearings. They operate under constant conditions, defined by a shaft speed of 1500 rpm, a radial force of 1000 N, and a load torque of 0.7 N·m [36]. The acceleration is initially measured at a high sampling frequency of 64 kHz. In this study, downsampling with a factor of four is applied before sample segmentation.

The PU dataset provides three sets of signals, including two current data and one vibration data. The original vibration signal is segmented into samples of 7921 data points each, resulting in a total of 5542 samples for the dataset used in this experiment. This paper uses the vibration signal, including three health states: normal, inner race fault, and outer race fault. Table III gives the sample sizes and corresponding labels. Similarly, five small sample datasets are prepared, and the detailed dataset information is shown in Table IV.

Finally, it is worth noting that all vibration signals are subjected to a unified preprocessing procedure before sample segmentation. Specifically, a physics-guided power spectral density (PSD) filter is applied to suppress low-frequency baseline noise and high-frequency mechanical interference, while preserving fault-related harmonics and sidebands. The

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TABLE II: Description of small sample datasets for CWRU.

Extraction ratio	Training set sample size	Validation set sample size	Test set sample size	Total sample size	Dataset name
10%	823	235	119	1177	C1
12%	989	282	142	1413	C2
14%	1153	329	166	1648	C3
16%	1318	376	190	1884	C4
18%	1484	424	212	2120	C5
20%	1648	471	236	2355	C6

cutoff frequency of the filter is set to $B_{\max} = 1000$ Hz. Subsequently, the input sample length is determined using the parameter selection strategy proposed in this study. A sliding window is then employed to expand the dataset, with a shift length of 500 for the CWRU dataset and 25 for the PU dataset [37]. This standardized preprocessing ensures consistency across experimental conditions and enhances the reliability of subsequent comparative analysis.

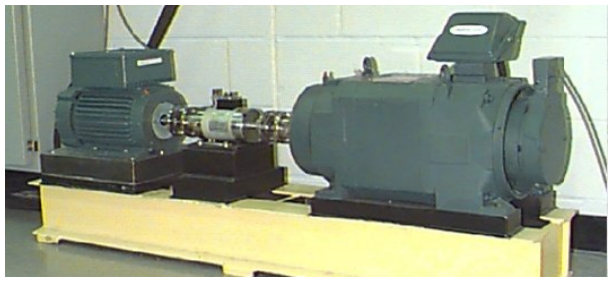


Fig. 3: CWRU bearing test bench [33].

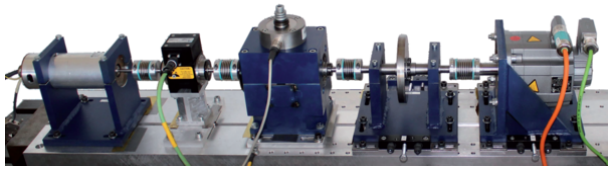


Fig. 4: PU bearing test bench [36].

TABLE I: Description of sample size and labels (CWRU dataset).

Condition	Sample size	Label
Normal	3357	0
Ball fault	2805	1
Inner race fault	2806	2
Outer race fault	2810	3

TABLE III: Description of sample size and labels (PU dataset).

Condition	Sample size	Label
Normal	1846	0
Inner race fault	1854	1
Outer race fault	1842	2

B. PG-Inception input length and size verification test

According to the method proposed in this paper, the PG-Inception network first requires the design of the input sample length and size based on the characteristics of the bearing vibration signal.

1) Input length: As discussed in Section II-B1, the sample input length of PG-Inception can be obtained by solving Equations (1)-(3), which represents the minimum length required to preserve adequate information for fault classification. For the CWRU bearing dataset, according to Equation (1), the minimum length under the first constraint is 224, while Equation (2) yields a minimum length of 5286 under the second constraint. For subsequent experiments, the smallest square number 5329 is selected as the sample input length above the minimum length. The same calculation method can be used to obtain the sample input length of the PU dataset to be 7921.

First, the CWRU bearing dataset is used to verify the effectiveness of the proposed method. To validate the calculated input length of 5329, samples with lengths of 1024, 2048, 4096, 6400, and 8100 are set for comparison. The experiments are conducted using the baseline Inception network with convolution kernel sizes of (3, 3) and (5, 5) for 30 epochs. As shown in Fig. 5, after the input length reaches 5329, the mean accuracy fluctuates within the range of 98% ± 0.5%, with a maximum value exceeding 99.5%, and increases with increasing length. The highest mean accuracy of 98.432% is achieved with an input length of 5329, and the standard deviation over 10 trials is 0.00826, indicating the highest stability. When the input length is 8100, the average accuracy is 98.249%, with a maximum of 99.85%, slightly higher than the lower limit of the input length calculated in this paper, but the input sample length increases by 51%. From the training efficiency analysis, the training time is 12.250s with an input length of 5329 and 14.840s with an input length of 8100, an increase of approximately 21%. Therefore, increasing the input sample length reduces the model's training efficiency. For input lengths of 4096 and 2048, the mean, median, and stability of the accuracy are not significantly different, indicating similar performance. When the input length is 1024, the average accuracy drops to 91.993%. Therefore, experiments demonstrate that, considering both accuracy and training efficiency, the proposed method calculates 5329 as the optimal input sample length. Similarly, for the PU dataset, 7921 is the optimal input sample length. Based on the experimental results of the two datasets, the proposed input length design method is proven effective. When using the CWRU and PU datasets with input lengths of 5329 and 7921 respectively, the samples contain sufficient fault information, thus enabling accurate classification.

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TABLE IV: Description of small sample datasets for PU.

Extraction ratio	Training set sample size	Validation set sample size	Test set sample size	Total sample size	Dataset name
44%	1706	487	245	2438	P1
46%	1784	509	256	2549	P2
48%	1861	532	267	2660	P3
50%	1939	554	278	2771	P4
52%	2016	576	289	2881	P5

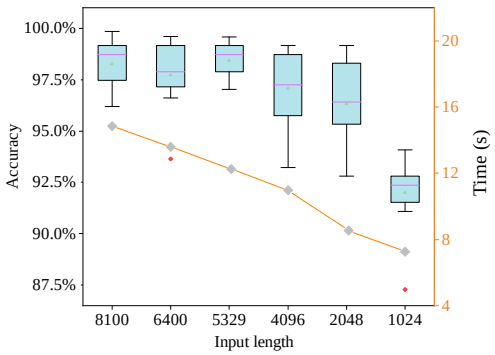


Fig. 5: Comparison of different input lengths.

2) Input size: As previously discussed, using a rectangular input sample improves the row-to-row feature representation in the reshaped input. Additionally, for the same number of data points, a rectangular input has fewer rows than a square input, reducing the likelihood of splitting measurements from the same fault across different rows [23]. After obtaining the optimal input length, this subsection will discuss its reshaped size.

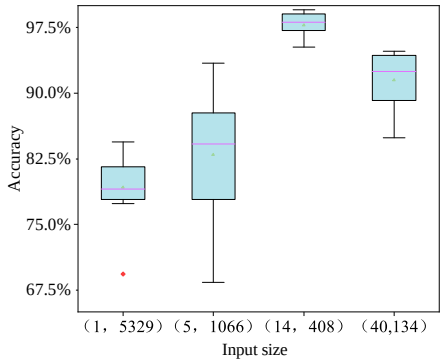


Fig. 6: Comparison of different input sizes.

The CWRU dataset is used to verify the input sample size design method. As illustrated in Fig. 2(b), the width and height of the input sample are determined by Equations (4) and (5), respectively. Following the calculations, the optimal sample size is determined to be (14, 408). For comparison, three datasets with sample sizes of (1, 5329), (5, 1066), and (40, 134) are prepared. To maintain the matrix shape, the insufficient positions in the sample are filled with zeros. The experiment is conducted using the baseline Inception network with convolution kernel sizes of (3, 3) and (5, 5). The experimental results are presented in Fig. 6. As the input size increases, the fault diagnosis performance of the PG-Inception network initially improves, then deteriorates, reaching its peak at a size of (14, 408). At this point, the average accuracy is 97.735%, with a standard deviation of just 0.01473. The performance is the worst when the input size is (1, 5329), with an average accuracy of only 79.197%. The result stability is the lowest when the input size is (5, 1066), with a standard deviation of 0.07279. These results validate the proposed method and identify (14, 408) as the optimal input size for this experiment. Similarly, for the PU dataset, optimal model performance is achieved with an input size of (13, 640). Based on the experiments with the two datasets above, the accuracy of the proposed input size design method is confirmed. The optimal input sizes for the PG-Inception network are (14, 408) for the CWRU dataset and (13, 640) for the PU dataset.

C. PG-Inception fault branch design verification test

In the PG-Inception network, the fault branch is designed using information from the bearing acceleration signal. This design involves two components: the convolution kernel size for each fault branch and the number of fault branches. As shown in Figs. 2(b) and 2(c), the process for the fault branch is specified, while the pooling branch and the branch with a convolution kernel size of (1,1) from the Inception network are retained.

1) Kernel size: According to Equations (7)-(9), different decaying ratios correspond to distinct kernel widths, extracting features from different scales. Based on the acceleration signal from the CWRU dataset, the convolution kernel widths for different decaying degrees can be calculated. The detailed information is given in Table V.

This study defines nine decaying ratios, ranging from 0.9 to 0.1 with a step size of 0.1, while the kernel height is set to the optimal value [23]. As shown in Table V, the three fault types correspond to specific convolution kernel widths at each attenuation level. Since the proposed method does not specify the kernel height, a single-branch structure is used to determine the optimal value. The experiment demonstrates that the optimal convolution kernel height is seven for the CWRU dataset [23]. Therefore, the height and width of the convolution kernel for the three fault branches are obtained. Using the dataset C5, 10 experiments are performed on the PG-Inception model for each attenuation level, with each experiment running for 30 epochs.

The experimental results are presented in Table VI and Fig. 7. Among the nine groups of PG-Inception with different convolution kernel size combinations, the model achieves the highest accuracy when the decay rate is defined as 0.1, with an average value of 97.735%. At this time, the model is also

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TABLE V: Kernel width with different decaying ratios (CWRU dataset).

Fault type	Decaying ratio								
	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Outer race fault	2	4	7	10	13	17	23	30	43
Inner race fault	2	5	8	11	16	20	27	35	52
Ball fault	6	14	21	27	40	52	70	97	134

TABLE VI: PG-Inception classification accuracy with different kernel widths (CWRU dataset).

ζ	Kernel size			Minimum	Maximum	Mean	Standard deviation
	Branch 1	Branch 2	Branch 3				
0.9	(7,2)	(7,2)	(7,6)	88.04%	94.34%	90.833%	0.01769
0.8	(7,4)	(7,5)	(7,14)	91.51%	97.64%	93.961%	0.0206
0.7	(7,7)	(7,8)	(7,21)	92.92%	98.11%	95.282%	0.02015
0.6	(7,10)	(7,11)	(7,27)	91.98%	96.7%	95.236%	0.01415
0.5	(7,13)	(7,16)	(7,40)	91.51%	99.06%	95.989%	0.02469
0.4	(7,17)	(7,20)	(7,52)	92.92%	98.58%	95.989%	0.01863
0.3	(7,23)	(7,27)	(7,70)	95.75%	99.53%	97.311%	0.01044
0.2	(7,30)	(7,35)	(7,97)	94.34%	98.11%	97.262%	0.01234
0.1	(7,43)	(7,52)	(7,134)	95.28%	99.53%	97.735%	0.01473

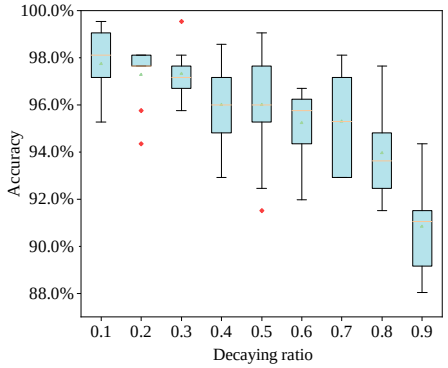


Fig. 7: Comparison of different kernel sizes.

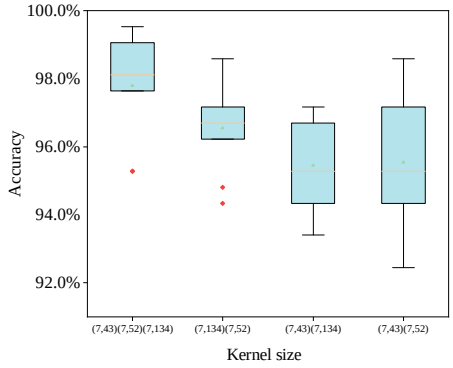


Fig. 8: Comparison of different parallel branches.

the most stable, with a standard deviation of only 0.01473. In terms of the average value, as the decay rate decreases from 0.9, the model's accuracy first increases, then decreases, and finally rises to the highest value. Based on the experimental results from the CWRU dataset, the optimal decay rate for the convolution kernel width design is determined to be 0.1, resulting in the following convolution kernel size combinations for the three fault branches: (7, 43), (7, 52), and (7, 134). Similarly, the convolution kernel sizes of the fault branches for the PU dataset are designed as (3, 51) and (3, 130).

2) Number of branches: According to Equation (6), except for one pooling branch and one convolution branch of size (1, 1), the number of parallel branches, specifically the fault branches, should correspond to the number of fault types. Under the experimental conditions, the bearing exhibits three fault types: inner race defect, outer race defect, and rolling element defect. Hence, the fault branch should consist of three branches. Building on the experimental findings from the previous section, the optimal decay rate for the convolution kernel width design is 0.1, leading to the following convolution size combinations for these three branches: (7, 43), (7, 52), and (7, 134). This section compares a two-branch structure using the following convolution size combinations: (7, 43) and (7, 52), (7, 43) and (7, 134), (7, 52) and (7, 134).

Experiments on the C5 dataset serve to validate the number of fault branches, as shown in Table VII and Fig. 8. First, a comparison of dual-branch structures shows the (7, 134) and (7, 43) kernel combination provides the best dual-branch performance, achieving a 96.7% mean accuracy and the lowest standard deviation (0.01314).

This optimal dual-branch structure is then compared to the proposed three-branch structure, which aligns with the three fault types. The three-branch model outperforms it in all key metrics, achieving a higher maximum (99.53%), minimum (95.28%), and mean accuracy (97.74%). Although the three-branch model's stability (std. 0.01473) is slightly lower than the best dual-branch pair, its superior accuracy confirms that performance is optimal when the number of fault branches matches the number of fault types. This conclusion is reinforced by experiments on the PU dataset, validating the effectiveness of the proposed physics-guided design methodology.

D. Accuracy and efficiency comparison

In the first two sections, the design of the PG-Inception network is completed with guidance from physical information.

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TABLE VII: PG-Inception classification accuracy with different number of branches (CWRU dataset).

Kernel size			Minimum	Maximum	Mean	Standard deviation
Branch 1	Branch 2	Branch 3				
(7,43)	(7,52)	/	92.45%	98.58%	95.33%	0.0203
(7,43)	(7,134)	/	93.40%	97.17%	95.33%	0.01323
(7,52)	(7,134)	/	94.34%	98.58%	96.70%	0.01314
(7,43)	(7,52)	(7,134)	95.28%	99.53%	97.74%	0.01473

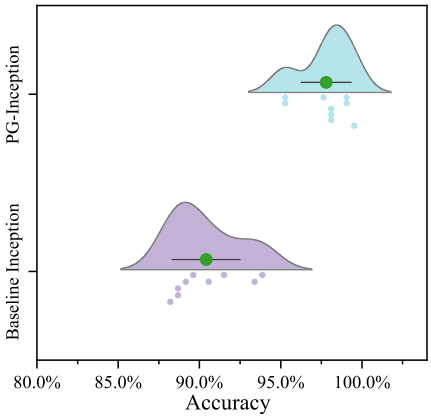


Fig. 9: Accuracy comparison for CWRU dataset.

To verify the superiority of this method, PG-Inception and baseline Inception are compared from multiple perspectives. First, the CWRU dataset is used to conduct a comparative experiment on the accuracy and efficiency of PG-Inception and baseline Inception. The experiment uses the C5 dataset and conducts 10 experiments in total, each lasting 30 epochs. In the original Inception structure proposed by Szegedy, the number of fault branches is two and the convolution kernels are all square-shaped [11]. To ensure that the number of fault branches in both baseline Inception and PG-Inception is the same in the comparative experiment, the design specifies that when baseline Inception has three fault branches, the convolution kernel sizes are (3, 3), (5, 5), and (7, 7).

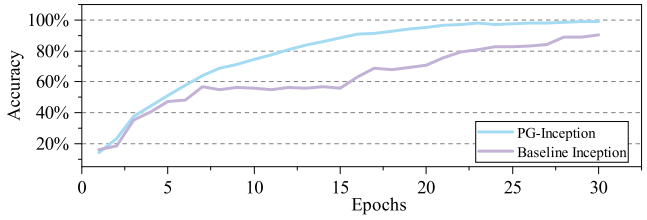


Fig. 10: Accuracy-epoch line plot for CWRU dataset.

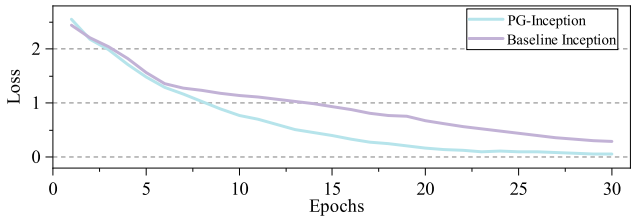


Fig. 11: Loss-epoch line plot for CWRU dataset.

As displayed in Table VIII and Fig. 9, the average accuracy of PG-Inception is 97.735%, which is 7.604% higher than that of baseline Inception. Even the minimum value of PG-Inception, 95.28%, exceeds the maximum value of baseline Inception, which is 93.87%. PG-Inception reveals superior performance in terms of maximum, minimum, mean, and standard deviation. The curves in Figs. 10 and 11 represent the average values from 10 experiments. PG-Inception converges significantly faster and more stably than the baseline. After sixth epoch, PG-Inception's accuracy rose from 50% to 90% in only 10 epochs, a milestone that took the baseline 25 epochs. PG-Inception achieved full convergence by the 22nd epoch, six epochs earlier than the baseline. In contrast, the baseline's accuracy stagnant between epochs 7-15 , and its loss curve shows inflection points, indicating an unstable process. Similarly, a comparative experiment is conducted using the PU and P4 datasets, and the results are presented in Table IX and Figs. 12, 13 and 14. Based on the results and analysis of the two comparative experiments, PG-Inception demonstrates superior diagnostic accuracy and efficiency compared to baseline Inception, confirming its superiority.

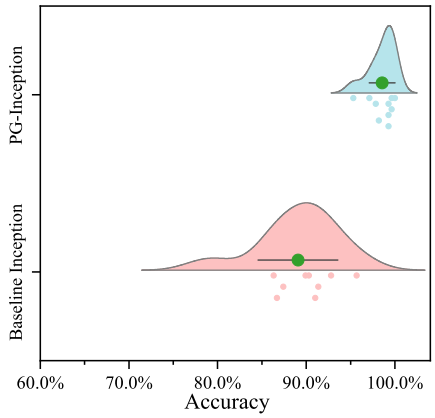


Fig. 12: Accuracy comparison for PU dataset.

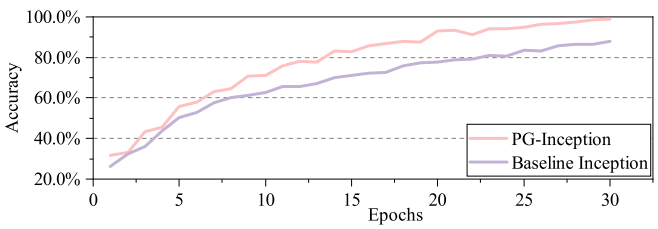


Fig. 13: Accuracy-epoch line plot for PU dataset.

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TABLE VIII: Classification accuracy with PG-Inception and baseline Inception (CWRU dataset).

Model	Minimum	Maximum	Mean	Standard deviation
Baseline Inception	88.21%	93.87%	90.331%	0.01993
PG-Inception	95.28%	99.53%	97.735%	0.01473

TABLE IX: Classification accuracy with PG-Inception and baseline Inception (PU dataset).

Model	Minimum	Maximum	Mean	Standard deviation
Baseline Inception	79.14%	95.68%	89.066%	0.04515
PG-Inception	95.32%	100.00%	98.56%	0.0146

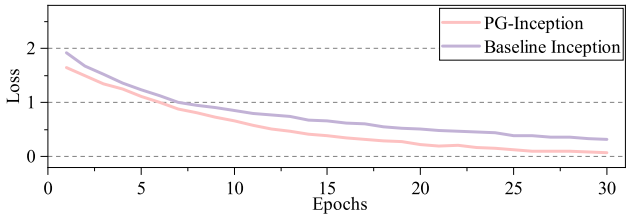


Fig. 14: Loss-epoch line plot for PU dataset.

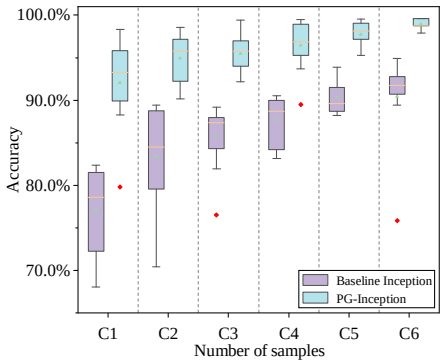


Fig. 15: Accuracy comparison under different small sample sizes for CWRU datase.

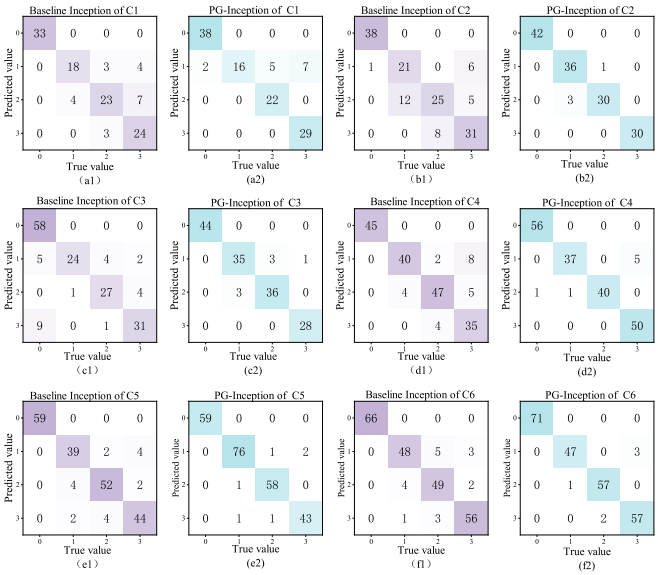


Fig. 16: Confusion matrix comparison for CWRU dataset.

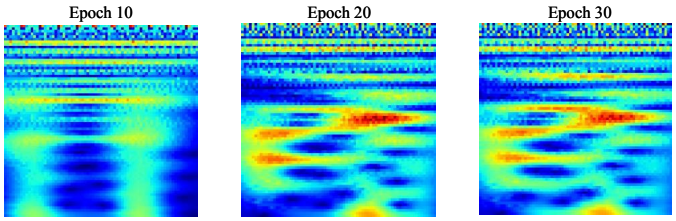
E. Performance comparison under different small samples

Due to the multi-scale parallel structure of the Inception network, it can extract features from different fields of view, making it highly effective under small sample conditions. This section compares the performance of the PG-Inception network with the baseline Inception network under various small sample conditions. Six groups of comparative experiments are first conducted using the CWRU bearing dataset.

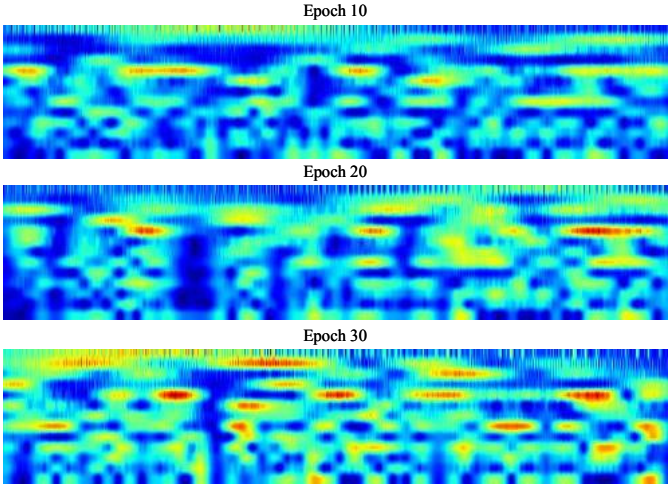
As Fig. 15 shows, PG-Inception outperforms the baseline Inception across all small-sample conditions, and its advantage widens as sample size decreases. While the accuracy gap is 8.559% on the C6 dataset, it increases to 14.955% on the C1 dataset. This superior robustness is highlighted by a key finding: PG-Inception on the smallest C1 dataset (with half the samples) still surpasses the baseline's accuracy on the largest C6 dataset. This demonstrates that integrating physical information significantly enhances the Inception network's performance in limited-data scenarios.

Fig. 16 presents the confusion matrix for each set of experimental results. Both PG-Inception and baseline Inception networks exhibit relatively high accuracy in identifying healthy states and rolling element faults, but both models tend to make more errors when identifying inner race and outer race faults. To further analyze the differences in feature extraction performance between the two models, their feature maps are examined. Fig. 17 illustrates the feature map performance of the baseline Inception model with kernel sizes of (3, 3) and (5, 5) using an input size of (73, 73), and the PG-Inception model with kernel sizes of (7, 43), (7, 52), and (7, 134) using an input size of (14, 408) on the C2 dataset. Fig. 17(a) shows the feature map of the baseline Inception model. As the training epoch increases, the feature map becomes more detailed. Starting from epoch 10, the feature map is relatively fuzzy, with activation values scattered across the signal. By epoch 30, the distribution of activation values becomes more concentrated, yet there are still noisy areas, and the activation values are insufficiently focused on fault-relevant regions, limiting the model's ability to capture global features effectively. In contrast, Fig. 17(b) presents the feature map of the PG-Inception model. Compared to the baseline, the PG-Inception model generates clearer and more focused feature representations at the same epoch. At epoch 30, the activation values of the PG-Inception feature map become more uniformly distributed and structured, with more prominent features in key regions. This

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(a) Feature map of baseline Inception with input size of (73, 73)



(b) Feature map of PG-Inception with input size of (14, 408)

Fig. 17: Comparison of feature maps in baseline Inception and PG-Inception.

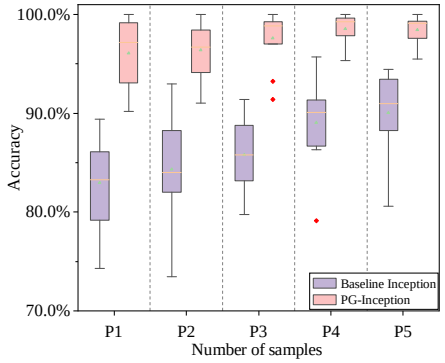


Fig. 18: Accuracy comparison under different small sample sizes (PU dataset).

improvement can be attributed to the physics-guided kernel sizes, which are derived from the fault characteristic frequency range and its harmonics. Specifically, the longer temporal kernels correspond to the characteristic vibration periods of bearing defects, enabling the network to effectively capture both periodic impulse components and their attenuation behavior. In contrast, the black-box kernels of the baseline model lack such correspondence, leading to less interpretable and less physically consistent feature responses. This indicates that PG-Inception's feature extraction ability improves progressively throughout the training process, leading to faster convergence.

The analysis confirms PG-Inception's superior feature extraction. Its feature maps reveal more concentrated activations,

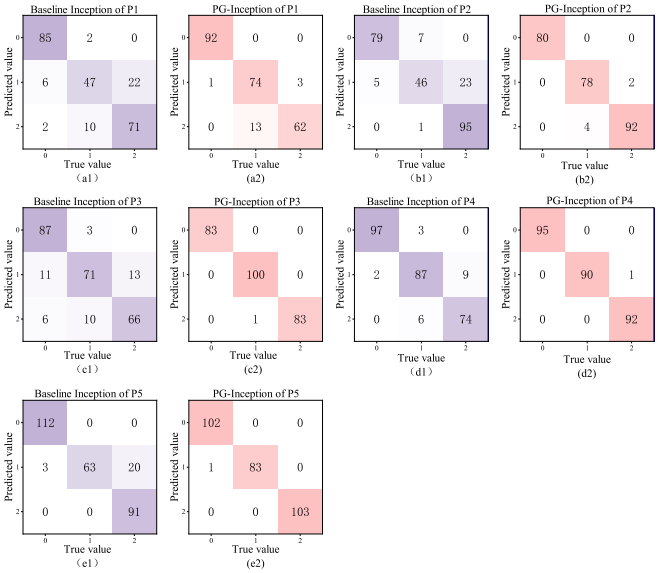


Fig. 19: Confusion matrix comparison for PU dataset.

less noise, and faster capture of key information, which enhances the Inception network's multi-scale recognition capabilities. This translates to significant improvements in small-sample scenarios. PG-Inception consistently outperforms the baseline, and its performance declines more slowly as sample size decreases; it even achieves higher accuracy with half the sample data. These findings, reinforced by the PU dataset experiments (see Figs. 18-19), clearly verify the superiority and interpretability of the proposed physics-guided method.

TABLE X: Comparison with PG-Inception and baseline Inception (CWRU dataset).

Model	Model running time	Model size	Number of parameters
Baseline Inception	52s	16.38 MB	4293194
PG-Inception	138s	17.86 MB	4682314

TABLE XI: Comparison with PG-Inception and baseline Inception (PU dataset).

Model	Model running time	Model size	Number of parameters
Baseline Inception	47s	16.32 MB	4278298
PG-Inception	80s	16.82 MB	4408602

F. Model size

Despite its superior accuracy, PG-Inception's practical deployment requires an evaluation of its computational cost. We compare model size, parameters, and runtime on a Google Colab L4-GPU platform, measuring runtime from a second pass to exclude dataset loading overhead.

On the C2 dataset (Table X), PG-Inception is 17.86 MB (1.48 MB larger) and has 389k more parameters than the baseline. This increase, attributed to the larger, physics-guided convolution kernels, results in a longer runtime (86s slower

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over 30 epochs). Data from the P2 dataset (Tablee XI) corroborates this finding. Consequently, the increased model size and computational time are clear limitations of the current PG-Inception model. Future work must focus on developing a lightweight version to enhance deployment efficiency.

IV. Discussion

A. Overall performance

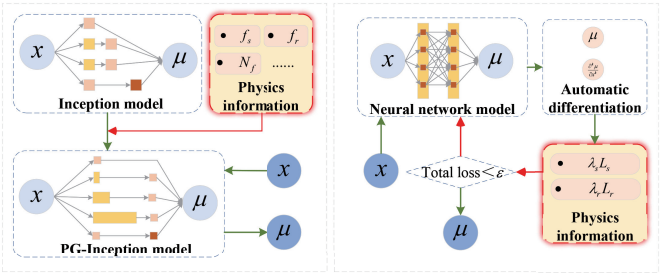
Recent advances in PIML have developed various approaches to integrate domain knowledge into neural networks, such as PINNs, physics-guided loss constraints, and hybrid feature embedding. These methods enhance the consistency between the model and physical laws. However, in practical applications, they require significant training time and computational cost because they rely on backward optimization during training for imposing physical constraints. Their effectiveness largely depends on the balance between loss weights and convergence dynamics. This paper proposes the PG-Inception method, which establishes a parameterized forward design approach. Based on bearing geometry parameters and operating condition information, it designs the input length, input size, convolutional kernel size, and number of branches of the Inception model. Therefore, this physics-to-structure mapping transforms implicit physical knowledge into explicit architectural rules, achieving higher design efficiency and interpretability compared to purely data-driven optimization. Experimental results on the CWRU and PU datasets demonstrate that the model designed using this method outperforms baseline-Inception in terms of accuracy and stability. In comparative experiments on the CWRU and PU datasets, PG-Inception achieved an accuracy improvement of 8.559%, from 90.384% to 98.943%, with a standard deviation reduction of 0.0052. The results demonstrate that the proposed method improves the performance of the Inception model. This is attributed to the proposed forward design method, which better matches the data and model. The designed input length provides sufficient fault information for the model; the input size preserves complete information; the designed convolutional kernel size provides the model with an appropriate field of view to perceive fault information; and the number of parallel branches provides suitable convolutional kernels for each fault type. When the training sample size is reduced from 1648 to 823, the accuracy of PG-Inception improves by 14.599% compared to baseline-Inception. This proves that under small sample conditions, the performance degrades more slowly as the sample size decreases. As can be seen from the feature map in the Fig. 17, the feature region resolution of PG-Inception is higher than that of baseline-Inception. This result also explains why the proposed method enhances the structural advantages of the Inception network in multi-scale recognition. The above experiments all demonstrate the effectiveness and superiority of this method.

B. Interpretability and few-shot

As mentioned in Section I, the method proposed in this paper can be regarded as a new implementation framework for PINNs. As shown in Fig. 20, compared with the current backward optimization approach that uses physical information to guide the design of loss functions, PG-Inception directly designs the parameters and structure of the model using physical information. This forward design approach improves the efficiency of using physical information, thereby enabling a more direct interpretation of the model's mapping rules. To further illustrate the advantages of this method in terms of interpretability, this paper conducts a simple comparative experiment to compare the performance of PG-Inception and PINN. The physics-informed loss of the PINN model is given by Equation (10) [38]:

$$\mathcal{L} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{FT}}\mathcal{L}_{\text{FT}}. \tag{10}$$

To improve the likelihood of PINN model convergence, the experimental dataset is conducted on C6, with 30 training epochs and 10 training runs. The experimental results are shown in Table XII. PG-Inception performs better in terms of accuracy. Fig. 21 shows the robustness test results [39]. The experimental results show that the accuracy of both models decreases as the SNR value decreases. When SNR = 3 dB, the accuracy of PG-Inception drops sharply, while the accuracy of PINN only decreases significantly when SN R= 0 dB. This is because PG-Inception did not consider the impact of noise during its design, focusing entirely on making the data and model more compatible, without paying attention to improving the robustness of Inception itself. Both experiments demonstrate the advantages of the proposed method in terms of interpretability.



(a) Forward design: PG-Inception (b) Backward optimization: PINN

Fig. 20: Flowchart of conceptual comparison.

Furthermore, the design framework of PG-Inception does not rely on prior knowledge of specific fault types. The network parameters are not derived from predefined fault labels, but from the inherent physical characteristics of the bearing. This formula based on physical principles also demonstrates that PG-Inception has good interpretability.

Due to the structural advantage of Inception's multi-scale recognition, this paper also compares the model performance of PG-Inception and baseline-Inception under small sample conditions. Meta-learning is also a widely used small sample learning method [39], [40], so a comparative experiment is also conducted [41]. The experimental results are shown in Fig. 22. The model performance of baseline-Inception itself under small sample conditions is inferior to that of Meta-learning. As the number of samples decreases, PG-Inception

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TABLE XII: Accuracy comparison with PG-Inception and PINN.

Model	Minimum	Maximum	Mean	Standard deviation
PINN	95.28%	99.53%	95.408%	0.01548
PG-Inception	97.88%	99.58%	98.943%	0.00609

effectively mitigates the rate of accuracy decline, and under the C1 dataset, PG-Inception still outperforms Meta-learning. These experiments demonstrate that this method effectively improves the performance of the Inception network under small sample conditions.

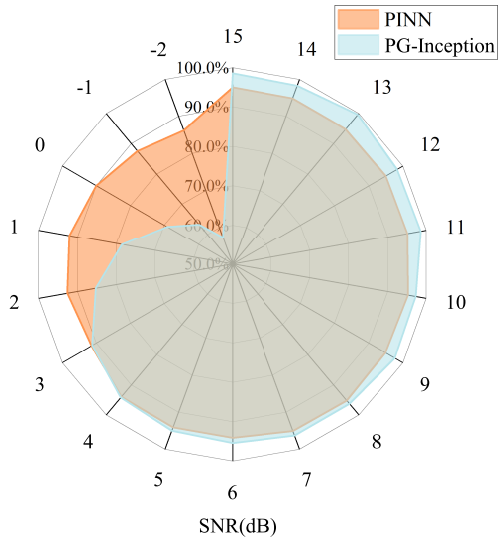


Fig. 21: Comparison with PG-Inception and PINN under different levels of noise interference.

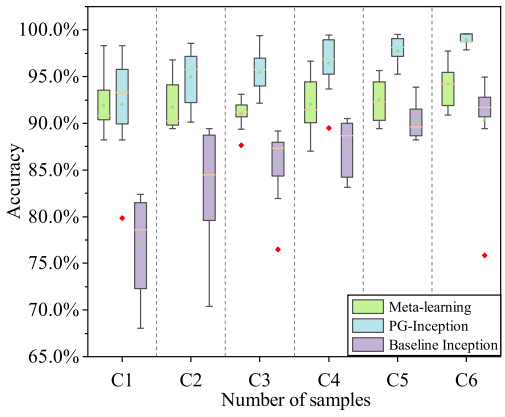


Fig. 22: Accuracy comparison under different small sample sizes.

C. Limitations and future directions

Although the proposed method's feasibility and advantages have been demonstrated, it possesses several limitations. As comparative results showed, robustness in low signal-to-noise ratio scenarios remains a key shortcoming. This section details these limitations to guide future research and development.

First, the framework's reliance on stationary conditions (constant speed and sampling rate) for FCF estimation is a key limitation. Under non-stationary or coupled-fault conditions, FCFs may shift or overlap, invalidating the static kernel and branch structure design. This can lead to errors and incomplete feature coverage. Future work should incorporate adaptive mechanisms to dynamically adjust network parameters based on operating state. Integrating speed observers or virtual tachometers could also enable accurate FCF tracking, extending PG-Inception's applicability to variable-speed scenarios.

Further limitations include the model's large size, which necessitates future work on lightweighting for efficient embedded deployment using techniques like network pruning or depthwise separable convolutions. The current framework, validated only on ball bearings, also requires broader verification. Future research should explore its applicability to other rotating components and diverse industrial scenarios, such as wind turbine and aircraft engine bearings. Finally, further optimization of architectural parameters (e.g., channels, pooling) and robustness enhancements via adversarial training are promising directions.

V. Conclusion

Integrating domain information into deep learning models can effectively improve training efficiency and interpretability. However, current PINNs implementation frameworks are mainly based on backward optimization, using physical information to constrain the loss function. This approach cannot directly interpret the high-dimensional mapping relationships of the network, thus limiting training efficiency.

This paper proposes PG-Inception, a forward-design approach based on physical information, providing a novel PINN implementation framework. Its feasibility and superiority are verified on the CWRU and PU bearing datasets. First, the input length and size are designed based on the fault cycle and frequency resolution, providing sufficient and complete fault information for the model. Second, the convolution kernel size is derived by fitting the decay envelope of the fault excitation using an exponential function, providing the model with an appropriate fault information perception field. Finally, the number of parallel branches is determined based on the number of fault types, providing appropriate convolution kernels for each fault. After constructing PG-Inception, a series of comparative experiments with baseline-Inception verify the feasibility and superiority of each parameter and structural design. This demonstrates that PG-Inception effectively improves model performance and interpretability.

Future research will focus on enhancing the robustness of the model based on physical information, enabling validation under a wider range of operating conditions. In particular, applying this method under non-stationary conditions remains a major challenge and a promising direction for further study.

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REFERENCES

[1] F. F. Diego, M. R. David, F. R. Oscar, and A. B. Amparo, "Automatic bearing fault diagnosis based on one-class ν-svm," *Computers & Industrial Engineering*, vol. 64, pp. 357–365, 2013.

[2] S. Zhang, S. B. Zhang, B. N. Wang, and T. G. Habetler, "Deep learning algorithms for bearing fault diagnostics—a comprehensive review," *IEEE Access*, vol. 8, pp. 29 857–29 881, 2020.

[3] X. S. Lou and K. A. Loparo, "Deep learning algorithms for bearing fault diagnostics—a comprehensive review," *IEEE Access*, vol. 8, pp. 29 857–29 881, 2020.

[4] A. S. Zhang, S. B. Li, Y. X. Cui, W. L. Yang, R. Z. Dong, and J. J. Hu, "Limited data rolling bearing fault diagnosis with few-shot learning," *IEEE Access*, vol. 7, pp. 110 895–110 904, 2019.

[5] X. Y. Liu, G. Chen, H. Wang, and X. K. Wei, "A siamese cnn-bilstm-based method for unbalance few-shot fault diagnosis of rolling bearings," *Measurement and Control*, vol. 57, pp. 551–565, 2024.

[6] J. S. Xie, J. Liu, T. Q. Ding, T. T. Wang, and T. J. Yu, "Self-attention metric learning based on multiscale feature fusion for few-shot fault diagnosis," *IEEE Sensors Journal*, vol. 23, pp. 19 771–19 782, 2023.

[7] D. H. Lee and J. P. Jeong, "Few-shot learning-based light-weight wdcnn model for bearing fault diagnosis in siamese network," *Sensors*, vol. 23, p. 6587, 2023.

[8] D. W. Ruan, L. Ma, Y. Y. Yang, J. P. Yan, and C. Gühmann, "Improvement by monte carlo for trajectory similarity-based rul prediction," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–11, 2024.

[9] D. W. Ruan, X. Z. Song, C. Gühmann, and J. P. Yan, "Collaborative optimization of cnn and gan for bearing fault diagnosis under unbalanced datasets," *Lubricants*, vol. 9, p. 105, 2021.

[10] L. Cheng, Z. An, Y. J. Guo, M. F. Ren, Z. L. Yang, and S. McLoone, "Mmfsl: A novel multimodal few-shot learning framework for fault diagnosis of industrial bearings," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–13, 2023.

[11] C. Szegedy, W. Liu, Y. Q. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," 2015, paper presented at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 1–9 June 2015.

[12] Z. B. Xu, Y. Wang, W. Xiong, and Z. G. Wang, "A novel attentional feature fusion with inception based on capsule network and application to the fault diagnosis of bearing with small data samples," *Machines*, vol. 10, p. 789, 2022.

[13] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, paper presented at the 32nd International Conference on Machine Learning, 7–9 Jul 2015.

[14] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," 2016, paper presented at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), June 2016.

[15] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," 2017, paper presented at the AAAI Conference on Artificial Intelligence, Feb 2017.

[16] Y. B. Liu, J. Li, Q. J. Li, and Q. Wang, "Transfer learning with inception resnet-based model for rolling bearing fault diagnosis," *Journal of Advanced Mechanical Design, Systems, and Manufacturing*, vol. 16, pp. JAMDSM0023–JAMDSM0023, 2022.

[17] J. Q. Tian and G. Bin, "A cross-domain intelligent fault diagnosis method based on feature transfer with improved inception resnet for rolling bearings under varying working condition," *Journal of Advanced Mechanical Design, Systems, and Manufacturing*, vol. 18, pp. JAMDSM0012–JAMDSM0012, 2024.

[18] Y. H. Wei, G. Wang, and J. Y. Wu, "Recognition of tool wear state based on inception-bilstm under few-shot," *Modular Machine Tool & Automatic Manufacturing Technique*, vol. 5, pp. 147–151, 2024.

[19] S. Zhang, F. Ye, B. N. Wang, and T. G. Habetler, "Few-shot bearing fault diagnosis based on model-agnostic meta-learning," *IEEE Transactions on Industry Applications*, vol. 57, pp. 4754–4764, 2021.

[20] R. Q. Yan, Z. G. Shang, Z. Y. Wang, W. G. Xu, Z. B. Zhao, S. B. Wang, and X. F. Chen, "Challenges and opportunities of xai in industrial intelligent diagnosis:priori-empowered," *JOURNAL OF MECHANICAL ENGINEERING*, vol. 60, p. 1, 2024.

[21] Z. M. Zhang and C. Sun, "Structural damage identification via physics-guided machine learning: a methodology integrating pattern recognition with finite element model updating," *Structural Health Monitoring*, vol. 20, pp. 1675–1688, 2020.

[22] X. F. Chen, M. Ma, Z. B. Zhao, Z. Zhai, and Z. Mao, "Physics-informed deep neural network for bearing prognosis with multisensory signals," *monitoring and diagnostics*, vol. 4, pp. 200–207, 2022.

[23] D. W. Ruan, J. Wang, J. P. Yan, and C. Gühmann, "Cnn parameter design based on fault signal analysis and its application in bearing fault diagnosis," *Advanced Engineering Informatics*, vol. 55, p. 101877, 2023.

[24] S. Shen, H. Lu, M. Sadoughi, C. Hu, V. Nemani, A. Thelen, K. Webster, M. Darr, J. Sidon, and S. Kenny, "A physics-informed deep learning approach for bearing fault detection," *Engineering Applications of Artificial Intelligence*, vol. 103, p. 104295, 2021.

[25] W. Q. Jiang, Y. Hong, B. T. Zhou, X. He, and C. Cheng, "A gan-based anomaly detection approach for imbalanced industrial time series," *IEEE Access*, vol. 7, pp. 143 608–143 619, 2019.

[26] P. F. Wen, Z. S. Ye, Y. Li, S. W. Chen, and S. Zhao, "Physics-informed neural networks for prognostics and health management of lithium-ion batteries," *IEEE Transactions on Intelligent Vehicles*, vol. 9, pp. 2276–2289, 2024.

[27] A. K. Barimah, O. P. Onu, O. Niculita, A. Cowell, and M. D., "Scalable data transformation models for physics-informed neural networks (pinns) in digital twin-enabled prognostics and health management (phm) applications," *computers*, vol. 4, p. 121, 2025.

[28] J. Y. Chai, H. Zeng, A. M. Li, and E. W. T. Ngai, "Deep learning in computer vision: A critical review of emerging techniques and application scenarios," *Machine Learning with Applications*, vol. 6, p. 100134, 2021.

[29] S. Wang, B. Kang, J. L. Ma, X. J. Zeng, M. M. Xiao, J. Guo, M. J. Cai, J. Y. Yang, Y. D. Li, X. F. Meng, and B. Xu, "A deep learning algorithm using ct images to screen for corona virus disease (covid-19)," *European Radiology*, vol. 31, pp. 6096–6104, 2021.

[30] X. L. Xia, C. Xu, and B. Nan, "Inception-v3 for flower classification," 2017, paper presented at the 2017 2nd International Conference on Image, Vision and Computing (ICIVC), 02-04 June 2017.

[31] C. Wang, D. L. Chen, L. Hao, X. B. Liu, Y. Zeng, and J. W. Chen, "Pulmonary image classification based on inception-v3 transfer learning model," *IEEE Access*, vol. 7, pp. 146 533–146 541, 2019.

[32] M. L. Richter, W. Byttner, U. Krumnack, A. Wiedenroth, L. Schallner, and J. Shenk, "(input) size matters for cnn classifiers," 2021, paper presented at the Artificial Neural Networks and Machine Learning – ICANN 2021, 07 Sep 2021.

[33] D. Neupane and J. W. Seok, "Bearing fault detection and diagnosis using case western reserve university dataset with deep learning approaches: A review," *IEEE Access*, vol. 8, pp. 93 155–93 178, 2020.

[34] S. Bakshi and S. Rajan, "Fall event detection system using inception-densenet inspired sparse siamese network," *IEEE Sensors Letters*, vol. 5, pp. 1–4, 2021.

[35] J. Y. Sun, S. Takeuchi, and L. Yamasaki, "Prototypical inception network with cross branch attention for time series classification," 2021, paper presented at the 2021 International Joint Conference on Neural Networks (IJCNN), 18-22 July 2021.

[36] C. Lessmeier, J. K. Kimotho, D. Zimmer, and W. Sextro, "Condition monitoring of bearing damage in electromechanical drive systems by using motor current signals of electric motors: A benchmark data set for data-driven classification," 2016, paper presented at the European Conference of the PHM Society 2016, 5 Jul 2016.

[37] D. W. Ruan, J. C. Liu, X. H. Wu, and J. P. Yan, "Design and implementation of a physics-guided psd filter for bearing vibration signal processing." *Measurement Science and Technology*, vol. 36, p. 066117, 2025.

[38] J. C. Liu, S. Q. Ye, D. W. Ruan, Z. Wu, Y. L. Qian, J. P. Yan, and C. Guhmann, "An optimized parallel physics-informed neural network for bearing dynamics modeling," 2025, pp. 1–7, paper presented at the2025 International Conference on Control, Automation and Diagnosis (ICCAD), 1–3 July 2025.

[39] X. Z. Li, G. P. Zhu, A. J. Hu, L. Xing, and L. Xiang, "A meta-learning method based on meta-feature enhancement for bearing fault identification under few-sample conditions." *Mechanical Systems and Signal Processing*, vol. 226, p. 112370, 2025.

[40] W. B. Song, D. Wu, W. M. Shen, and B. Boulet, "Early fault detection for rolling bearings: A meta-learning approach." *IET Collaborative Intelligent Manufacturing*, vol. 6, p. e12103, 2024.

[41] C. C. Yu, Y. Q. Ning, Y. Qin, W. J. Su, and X. Zhao, "Multi-label fault diagnosis of rolling bearing based on meta-learning." *Neural Computing and Applications*, vol. 33, pp. 5393–5407, 2021.